

electron/workspace-history.cjs

Kind: added · Baseline: a9c34bfb03fe · Target: NovaDiff · Generated: 2026-05-20T04:33:10.403Z

Overview

A new module `electron/workspace-history.cjs` is added to the Electron side of the repo. It implements snapshotting of Git commits, indexing of workspace history, and a refresh routine that re-reads the log and re-creates missing snapshots.

Key changes

- **Imports:** added `fsp`, `fssync`, `path`, `runGit/tryRunGit`, and `upsertWorkspace/getWorkspace` (lines 3-6, 33).
- **Utility helpers:** `removeDirSafe` (lines 8-16) and `isUsableSnapshotDir` (lines 18-32) guard filesystem operations.
- **Commit enumeration:** `listCommitsOldestFirst` (lines 35-58) runs `git log` and returns an array of commit objects.
- **Snapshot creation:** `snapshotCommit` (lines 61-78) materializes a detached worktree, validates it, and returns the path.
- **Single-commit snapshot:** `ensureCommitSnapshot` (lines 80-98) ties a snapshot to a workspace record via `upsertWorkspace`.
- **Full history indexing:** `indexWorkspaceHistory` (lines 100-183) walks all commits, creates snapshots, writes metadata, and updates workspace state.
- **Refresh logic:** `refreshWorkspaceHistory` (lines 188-291) optionally fetches remote refs, re-reads the log, re-uses existing snapshots, and updates the workspace.
- **Exports:** all public functions are exported (lines 293-300).

Impact

- **New API surface:** callers can now trigger history indexing or refresh via the exported functions.
- **Filesystem side-effects:** the module creates/clears directories under `workspace.dataDir`, which may affect disk usage and performance.
- **Git dependency:** relies on `git` being available; errors are surfaced through `runGit / tryRunGit`.
- **State persistence:** uses `upsertWorkspace / getWorkspace`, so existing workspace records must be compatible with the new `commits` structure.

Risks & follow-ups

- Verify that `workspace.dataDir` paths are correctly resolved on all OSes; path handling is critical.
- Ensure `runGit` and `tryRunGit` correctly propagate errors; missing Git binaries could crash the process.

- Performance regression: snapshotting every commit can be expensive; benchmark on large repos.
- Test that `refreshWorkspaceHistory` correctly re-uses existing snapshots and does not overwrite them inadvertently.